

Combined Impact of Health Behaviours and Mortality in Men and Women: The EPIC-Norfolk Prospective Population Study

Background There is overwhelming evidence that behavioural factors influence health, but their combined impact on the general population is less well documented. We aimed to quantify the potential combined impact of four health behaviours on mortality in men and women living in the general community. **Methods and Findings** We examined the prospective relationship between lifestyle and mortality in a prospective population study of 20,244 men and women aged 45–79 y with no known cardiovascular disease or cancer at baseline survey in 1993–1997, living in the general community in the United Kingdom, and followed up to 2006. Participants scored one point for each health behaviour: current non-smoking, not physically inactive, moderate alcohol intake (1–14 units a week) and plasma vitamin C >50 mmol/l indicating fruit and vegetable intake of at least five servings a day, for a total score ranging from zero to four. After an average 11 y follow-up, the age-, sex-, body mass-, and social class-adjusted relative risks (95% confidence intervals) for all-cause mortality (1,987 deaths) for men and women who had three, two, one, and zero compared to four health behaviours were respectively, 1.39 (1.21–1.60), 1.95 (1.70–2.25), 2.52 (2.13–3.00), and 4.04 (2.95–5.54) $p < 0.001$ trend. The relationships were consistent in subgroups stratified by sex, age, body mass index, and social class, and after excluding deaths within 2 y. The trends were strongest for cardiovascular causes. The mortality risk for those with four compared to zero health behaviours was equivalent to being 14 y younger in chronological age. **Conclusions** Four health behaviours combined predict a 4-fold difference in total mortality in men and women, with an estimated impact equivalent to 14 y in chronological age.

Editors' Summary Background. Every day, or so it seems, new research shows that some aspect of lifestyle—physical activity, diet, alcohol consumption, and so on—affects health and longevity. For the person in the street, all this information is confusing. What is a healthy diet, for example? Although there are some common themes such as the benefit of eating plenty of fruit and vegetables, the details often differ between studies. And exactly how much physical activity is needed to improve health? Is a gentle daily walk sufficient or simply a stepping stone to doing enough exercise to make a real difference? The situation with alcohol consumption is equally confusing. Small amounts of alcohol apparently improve health but large amounts are harmful. As a result, it can be hard for public-health officials to find effective ways to encourage the behavioral changes that the scientific evidence suggests might influence the health of populations. **Why Was This Study Done?** There is another factor that is hindering official attempts to provide healthy lifestyle advice to the public. Although there is overwhelming evidence that individual behavioral factors influence health, there is very little information about their combined impact. If the combination of several small differences in lifestyle could be shown to have a marked effect on the health of populations, it might be easier to persuade people to make behavioral changes to improve their health, particularly if those changes

were simple and relatively easy to achieve. In this study, which forms part of the European Prospective Investigation into Cancer and Nutrition (EPIC), the researchers have examined the relationship between lifestyle and the risk of dying using a health behavior score based on four simply defined behaviors—smoking, physical activity, alcohol drinking, and fruit and vegetable intake. What Did the Researchers Do and Find? Between 1993 and 1997, about 20,000 men and women aged 45–79 living in Norfolk UK, none of whom had cancer or cardiovascular disease (heart or circulation problems), completed a health and lifestyle questionnaire, had a health examination, and had their blood vitamin C level measured as part of the EPIC-Norfolk study. A health behavior score of between 0 and 4 was calculated for each participant by giving one point for each of the following healthy behaviors: current non-smoking, not physically inactive (physical inactivity was defined as having a sedentary job and doing no recreational exercise), moderate alcohol intake (1–14 units a week; a unit of alcohol is half a pint of beer, a glass of wine, or a shot of spirit), and a blood vitamin C level consistent with a fruit and vegetable intake of at least five servings a day. Deaths among the participants were then recorded until 2006. After allowing for other factors that might have affected their likelihood of dying (for example, age), people with a health behavior score of 0 were four times as likely to have died (in particular, from cardiovascular disease) than those with a score of 4. People with a score of 2 were twice as likely to have died. What Do These Findings Mean? These findings indicate that the combination of four simply defined health behaviors predicts a 4-fold difference in the risk of dying over an average period of 11 years for middle-aged and older people. They also show that the risk of death (particularly from cardiovascular disease) decreases as the number of positive health behaviors increase. Finally, they can be used to calculate that a person with a health score of 0 has the same risk of dying as a person with a health score of 4 who is 14 years older. These findings need to be confirmed in other populations and extended to an analysis of how these combined health behaviors affect the quality of life as well as the risk of death. Nevertheless, they strongly suggest that modest and achievable lifestyle changes could have a marked effect on the health of populations. Armed with this information, public-health officials should now be in a better position to encourage behavior changes likely to improve the health of middle-aged and older people. Additional Information. Please access these Web sites via the online version of this summary at <http://dx.doi.org/10.1371/journal.pmed.0050012>.